

LAB Assignment 08

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Table

flight_id	airline	source	destination	price	duration
1	Indigo	Delhi	Mumbai	5000	2
2	AirIndia	Kolkata	Chennai	6500	3
3	SpiceJet	Mumbai	Bangalore	4000	1.5
4	Vistara	Delhi	Chennai	7000	3
5	GoFirst	Mumbai	Delhi	4800	2
6	Indigo	Bangalore	Kolkata	5500	2.5

Table

booking id	passenger name	flight id	city	seats	total fare
101	Aarav	1	Delhi	1	5000
102	Meera	2	Kolkata	2	13000
103	Rohan	3	Mumbai	1	4000
104	Sneha	4	Delhi	3	21000
105	Kunal	5	Mumbai	2	9600
106	Priya	6	Bangalore	1	5500

QUESTIONS

Q1. Display all flights.

Query:**db.flights.find()**

Output:

_id	flight_id	airline	source	destination	price	duration
69d0b20a747480...	1	Indigo	Delhi	Mumbai	5000	2
69d0b20a747480...	2	AirIndia	Kolkata	Chennai	6500	3
69d0b20a747480...	3	SpiceJet	Mumbai	Bangalore	4000	1.5
69d0b20a747480...	4	Vistara	Delhi	Chennai	7000	3
69d0b20a747480...	5	GoFirst	Mumbai	Delhi	4800	2
69d0b20a747480...	6	Indigo	Bangalore	Kolkata	5500	2.5

Q2. Display passenger_name and city from bookings.

Query: db.bookings.find({}, { passenger_name: 1, city: 1, _id: 0 })

Output:

passenger_name	city
Aarav	Delhi
Meera	Kolkata
Rohan	Mumbai
Sneha	Delhi
Kunal	Mumbai
Priya	Bangalore

Q3. Find flights where source is Delhi.

Query: db.flights.find({ source: "Delhi" })

Output:

_id	flight_id	airline	source	destination	price	duration
69d0b20a747480...	1	Indigo	Delhi	Mumbai	5000	2
69d0b20a747480...	4	Vistara	Delhi	Chennai	7000	3

Q4. Find bookings where seats = 1.

Query: db.bookings.find({ seats: 1 })

Output:

_id	booking_id	passenger_name	flight_id	city	seats	total_fare
69d0b219747480...	101	Aarav	1	Delhi	1	5000
69d0b219747480...	103	Rohan	3	Mumbai	1	4000
69d0b219747480...	106	Priya	6	Bangalore	1	5500

Q5. Sort flights by price in ascending order.

Query: db.flights.find().sort({ price: 1 })

Output:

_id	flight_id	airline	source	destination	price	duration
69d0b20a747480...	3	SpiceJet	Mumbai	Bangalore	4000	1.5
69d0b20a747480...	5	GoFirst	Mumbai	Delhi	4800	2
69d0b20a747480...	1	Indigo	Delhi	Mumbai	5000	2
69d0b20a747480...	6	Indigo	Bangalore	Kolkata	5500	2.5
69d0b20a747480...	2	AirIndia	Kolkata	Chennai	6500	3
69d0b20a747480...	4	Vistara	Delhi	Chennai	7000	3

Q6. Display top 2 bookings with highest total_fare.

Query: `db.bookings.find().sort({ total_fare: -1 }).limit(2)`

Output:

_id	booking_id	passenger_name	flight_id	city	seats	total_fare
69d0b219747480...	104	Sneha	4	Delhi	3	21000
69d0b219747480...	102	Meera	2	Kolkata	2	13000

Q7. Find flights where price is greater than 5000.

Query: `db.flights.find({ price: { $gt: 5000 } })`

Output:

_id	flight_id	airline	source	destination	price	duration
69d0b20a747480...	2	AirIndia	Kolkata	Chennai	6500	3
69d0b20a747480...	4	Vistara	Delhi	Chennai	7000	3
69d0b20a747480...	6	Indigo	Bangalore	Kolkata	5500	2.5

Q8. Display flights where destination is not Chennai.

Query: `db.flights.find({ destination: { $ne: "Chennai" } })`

Output:

_id	flight_id	airline	source	destination	price	duration
69d0b20a747480...	1	Indigo	Delhi	Mumbai	5000	2
69d0b20a747480...	3	SpiceJet	Mumbai	Bangalore	4000	1.5
69d0b20a747480...	5	GoFirst	Mumbai	Delhi	4800	2
69d0b20a747480...	6	Indigo	Bangalore	Kolkata	5500	2.5

Q9. Find bookings where seats are between 1 and 2.

Query: `db.bookings.find({ seats: { $gte: 1, $lte: 2 } })`

Output:

_id	booking_id	passenger_name	flight_id	city	seats	total_fare
69d0b219747480...	101	Aarav	1	Delhi	1	5000
69d0b219747480...	102	Meera	2	Kolkata	2	13000
69d0b219747480...	103	Rohan	3	Mumbai	1	4000
69d0b219747480...	105	Kunal	5	Mumbai	2	9600
69d0b219747480...	106	Priya	6	Bangalore	1	5500

Q10. Update price by +500 for flights of airline "Indigo".

Query: `db.flights.updateMany({ airline: "Indigo" }, { $inc: { price: 500 } })`

Output: {acknowledged: true,

insertedId: null,

matchedCount: 2,

modifiedCount: 2,

upsertedCount: 0}

Q11. Delete bookings where seats are greater than 2.

Query: `db.bookings.deleteMany({ seats: { $gt: 2 } })`

Output:

acknowledged	deletedCount
 true	 1.0

Q12. Find total fare collected city-wise. (Aggregation)

Query: `db.bookings.aggregate([ { $group: { _id: "$city", totalFare: { $sum: "$total_fare" } } } ])`

Output:

_id	totalFare
 Kolkata	 13000
 Delhi	 5000
 Mumbai	 13600
 Bangalore	 5500

Q13. Find average flight price per airline. (Aggregation)

Query: `db.flights.aggregate([ { $group: { _id: "$airline", avgPrice: { $avg: "$price" } } } ])`

Output:

_id	avgPrice
 Vistara	 7000.0
 SpiceJet	 4000.0
 Indigo	 5750.0
 AirIndia	 6500.0
 GoFirst	 4800.0

Q14. Display airlines where average price is greater than 5500. (Aggregation + filter)

Query: `db.flights.aggregate([{$group: {_id: "$airline", avgPrice: { $avg: "$price" } }}, {$match: { avgPrice: { $gt: 5500 } } }])`

Output:

_id	avgPrice
AirIndia	6500.0
Indigo	5750.0
Vistara	7000.0

Q15. Find maximum total_fare among bookings for each city. (Aggregation)

Query: `db.bookings.aggregate([{$group: {_id: "$city", maxFare: { $max: "$total_fare" } } }])`

Output:

_id	maxFare
Bangalore	5500
Kolkata	13000
Delhi	5000
Mumbai	9600

Final output:

Flights table:

_id	flight_id	airline	source	destination	price	duration
69d0b20a747480...	1	Indigo	Delhi	Mumbai	5500	2
69d0b20a747480...	2	AirIndia	Kolkata	Chennai	6500	3
69d0b20a747480...	3	SpiceJet	Mumbai	Bangalore	4000	1.5
69d0b20a747480...	4	Vistara	Delhi	Chennai	7000	3
69d0b20a747480...	5	GoFirst	Mumbai	Delhi	4800	2
69d0b20a747480...	6	Indigo	Bangalore	Kolkata	6000	2.5

Bookings table:

_id	booking_id	passenger_name	flight_id	city	seats	total_fare
 69d0b219747480...	 101	 Aarav	 1	 Delhi	 1	 5000
 69d0b219747480...	 102	 Meera	 2	 Kolkata	 2	 13000
 69d0b219747480...	 103	 Rohan	 3	 Mumbai	 1	 4000
 69d0b219747480...	 105	 Kunal	 5	 Mumbai	 2	 9600
 69d0b219747480...	 106	 Priya	 6	 Bangalore	 1	 5500